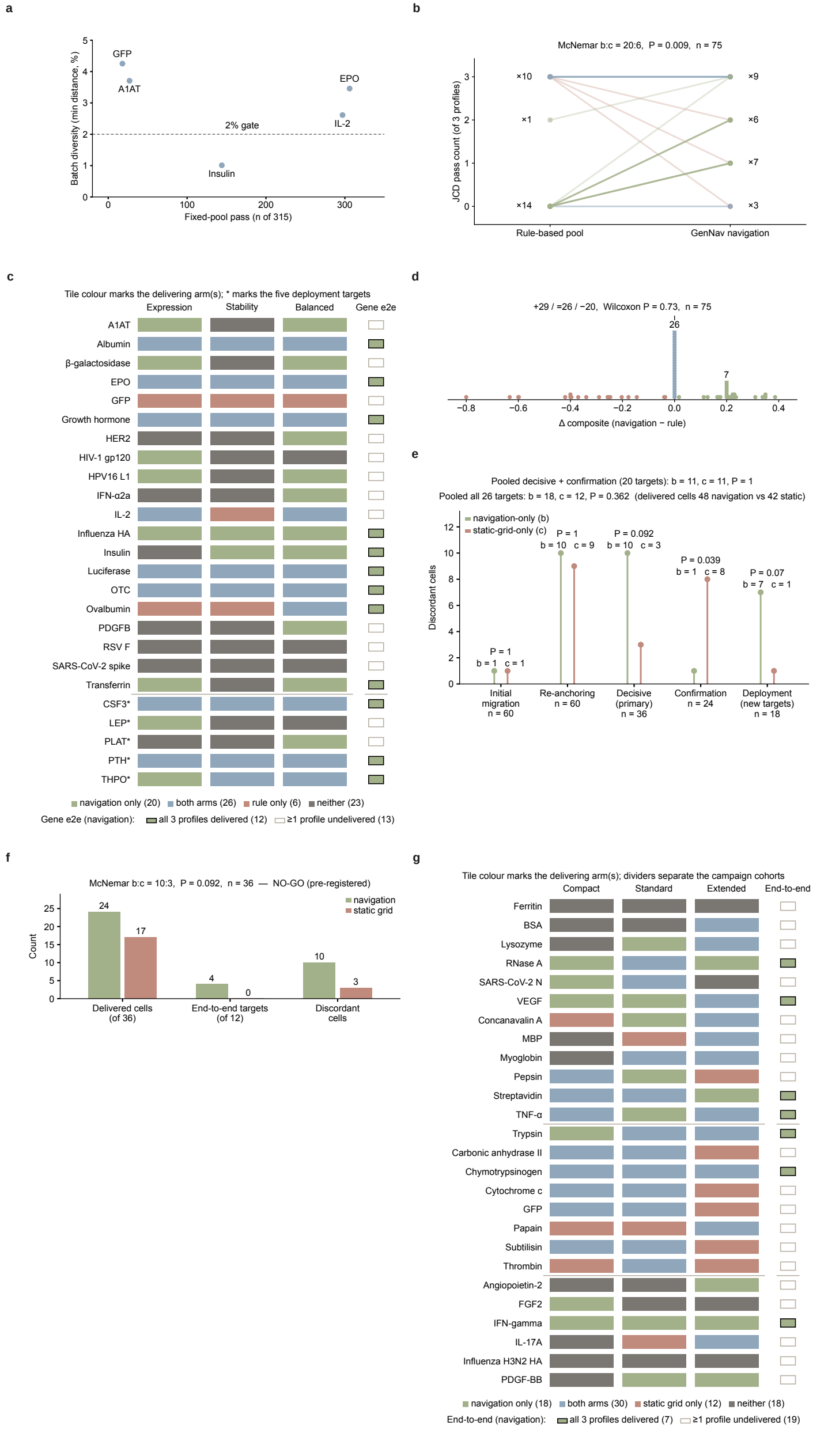

a
Batch diversity (min distance, %)
Fixed-pool pass (n of 315)
GFP
A1AT
EPO
IL-2
Insulin
2% gate
b
McNemar b:c = 20:6, P = 0.009, n = 75
JCD pass count (of 3 profiles)
Rule-based pool
GenNav navigation
×10
×1
×14
×9
×6
×7
×3
c
Tile colour marks the delivering arm(s); * marks the five deployment targets
Expression
Stability
Balanced
Gene e2e
A1AT
Albumin
β-galactosidase
EPO
GFP
Growth hormone
HER2
HIV-1 gp120
HPV16 L1
IFN-α2a
IL-2
Influenza HA
Insulin
Luciferase
OTC
Ovalbumin
PDGFB
RSV F
SARS-CoV-2 spike
Transferrin
CSF3*
LEP*
PLAT*
PTH*
THPO*
navigation only (20)
both arms (26)
rule only (6)
neither (23)
Gene e2e (navigation):
all 3 profiles delivered (12)
≥1 profile undelivered (13)
d
+29 / =26 / −20, Wilcoxon P = 0.73, n = 75
Δ composite (navigation − rule)
e
Pooled decisive + confirmation (20 targets): b = 11, c = 11, P = 1
Pooled all 26 targets: b = 18, c = 12, P = 0.362 (delivered cells 48 navigation vs 42 static)
navigation-only (b)
static-grid-only (c)
Discordant cells
P = 1
b = 1 c = 1
P = 1
b = 10 c = 9
P = 0.092
b = 10 c = 3
P = 0.039
b = 1 c = 8
P = 0.07
b = 7 c = 1
Initial migration n = 60
Re-anchoring n = 60
Decisive (primary) n = 36
Confirmation n = 24
Deployment (new targets) n = 18
f
McNemar b:c = 10:3, P = 0.092, n = 36 — NO-GO (pre-registered)
navigation
static grid
Count
Delivered cells (of 36)
End-to-end targets (of 12)
Discordant cells
g
Tile colour marks the delivering arm(s); dividers separate the campaign cohorts
Compact
Standard
Extended
End-to-end
Ferritin
BSA
Lysozyme
RNase A
SARS-CoV-2 N
VEGF
Concanavalin A
MBP
Myoglobin
Pepsin
Streptavidin
TNF-α
Trypsin
Carbonic anhydrase II
Chymotrypsinogen
Cytochrome c
GFP
Papain
Subtilisin
Thrombin
Angiopoietin-2
FGF2
IFN-gamma
IL-17A
Influenza H3N2 HA
PDGF-BB
navigation only (18)
both arms (30)
static grid only (12)
neither (18)
End-to-end (navigation):
all 3 profiles delivered (7)
≥1 profile undelivered (19)